

Phase 2 Verification and Validation									
Project / System Prepared By Version Date Approved By	Elevator System			Phase 2 expands verification and validation to the user experience/client program, database repository, diagnostic reporting, server authorization/logging, floor announcements, Sabbath Mode, and queue handling. The following items align with the Phase 2 project requirements and are intended to be customized as implementation details are finalized.					
	L Ebert								
	1.3 Jul 20, 2026								
	O Kipp								
Summary									
Total Tests	Passed	Failed	Blocked	Not Run		Completion %			
16	15	0	1	0		94%			
ID	Category	Sub Item	Verification Method	Expected Result	Test Procedure	Acceptance Criteria	Result	Validator / Date	Evidence / Notes
P2-VV-001	UX/ Client program	Front End GUI can control elevator	Functional test / observation	Front End GUI successfully sends elevator control commands and the elevator responds as intended.	1) Log into the client/front-end interface. 2) Issue elevator control command from GUI. 3) Observe elevator response and UI feedback. 4) Record result and evidence.	Pass when GUI command is accepted, transmitted, and reflected by elevator operation/UI status without unexpected errors.	Blocked		Unclear final requirement, GUI can control elevator but only in specific situations
P2-VV-002	UX/ Client program	User Action Buttons Implemented	Functional test / observation	All required user action buttons are present, labelled, and functional in the front-end GUI.	1) Open GUI. 2) Verify each required action button is visible. 3) Click each button. 4) Confirm intended UI/system response.	Pass when every required action button is implemented and produces the intended action.	Pass	OK/ 07-17-26	See video in Debrief 6 "Phase 2.1"
P2-VV-003	UX/ Client program	Indicators implemented	Functional test / observation	GUI indicators display correct elevator/client status information.	1) Open GUI. 2) Trigger or observe status changes. 3) Compare displayed indicators against expected system state/database values.	Pass when indicators are visible and match the actual system state.	Pass	BG/ 07-20-26	
P2-VV-004	UX/ Client program	User action buttons make changes to UI that are reflected in the Database	Functional test / observation	User action button changes update the UI and are written to the database.	1) Press a GUI action button. 2) Observe UI change. 3) Inspect database record/value. 4) Confirm UI and DB are consistent.	Pass when the UI updates and the corresponding database field records the same action/state.	Pass	OK/ 07-17-26	See video in Debrief 6
P2-VV-005	UX/ Client program	User indicator buttons change	Functional test / observation	User indicator buttons change state in response to user/system actions.	1) Trigger each indicator button condition. 2) Observe button state/colour/text change. 3) Compare against expected behaviour.	Pass when each indicator button changes state correctly and consistently.	Pass	BG/ 07-20-26	
P2-VV-006	Database Repository	DB set up and accepting values/commands from website	Functional test / observation	Database is configured and accepts valid values/commands submitted from the website.	1) Submit valid command/value from website. 2) Inspect database table/record. 3) Confirm stored value and timestamp/identifier as applicable.	Pass when valid website submissions are saved correctly without database errors.	Pass	OK/ 07-17-26	See video Phase2.2
P2-VV-007	Database Repository	DB has fields for logging information placed on the CAN bus	Functional test / observation	Database schema includes fields needed to log CAN bus information.	1) Review database schema. 2) Generate/observe CAN bus message data. 3) Confirm relevant fields can store CAN ID, message/data, source, timestamp, and status as applicable.	Pass when CAN bus logging fields exist and accept representative CAN bus log entries.	Pass	OK/ 07-17-26	See video Phase2.2
P2-VV-008	Diagnostic Reports	Produced based on content in DB	Functional test / observation	Diagnostic reports are generated from database content.	1) Populate database with known sample records. 2) Generate diagnostic report. 3) Compare report contents against DB values.	Pass when report output accurately reflects the selected database records.	Pass	OK/ 07-17-26	See video Phase2.3
P2-VV-009	Diagnostic Reports	Reports displayed on webpage	Functional test / observation	Diagnostic reports are displayed correctly on the webpage.	1) Navigate to report page. 2) Generate or load a report. 3) Verify report is visible, readable, and matches expected data.	Pass when report appears on the webpage with complete and accurate information.	Pass	OK/ 07-17-26	see video phase 2.3
P2-VV-010	Server Program	Users are authorized/ authenticated based on username/password	Functional test / observation	Server authenticates users using username/password and distinguishes authorized users.	1) Attempt login with valid credentials. 2) Attempt login with invalid credentials. 3) Verify access control outcome.	Pass when valid users are granted access and invalid users are denied access.	Pass	OK/ 07-17-26	see video phase 2.4
P2-VV-011	Server Program	Server program provides logging data/accepts only from Authorized users	Functional test / observation	Server provides logging data and accepts requests only from authorized users.	1) Log in as authorized user and request/submit logging data. 2) Repeat as unauthorized/invalid user. 3) Compare access results.	Pass when authorized users can access/submit logging data and unauthorized users are blocked.	Pass	OK/ 07-17-26	see video phase 2.4
P2-VV-012	Floor Announcements	Implement a server program for announcing the floor numbers	Functional test / observation	Server program produces floor number announcement events/messages.	1) Start server announcement function. 2) Simulate or run floor change events. 3) Verify announcement trigger/output for current/next floor.	Pass when server generates correct floor announcement output for each applicable floor event.	Pass	BG/ 07-27-26	
P2-VV-013	Floor Announcements	Speaker on elevator used to announce the current/next floor	Functional test / observation	Elevator speaker announces the current or next floor audibly.	1) Operate elevator through floor transitions. 2) Listen for speaker announcement. 3) Confirm announced floor matches current/next floor.	Pass when speaker announcement is audible and matches the correct floor.	Pass	BG/ 07-27-26	
P2-VV-014	Sabbath Mode	Users operate the elevator without having to push buttons	Functional test / observation	Sabbath Mode allows elevator operation without user button presses.	1) Enable Sabbath Mode. 2) Observe elevator operation sequence. 3) Confirm users can ride without pressing call or floor buttons.	Pass when elevator services floors according to Sabbath Mode behaviour without user button input.	Pass	BG/ 07-27-26	
P2-VV-015	Queues	Queues implemented for GUI inputs	Functional test / observation	GUI input commands are placed into and processed from an input queue.	1) Submit multiple GUI commands. 2) Inspect queue behaviour/order. 3) Confirm commands are processed in expected sequence.	Pass when GUI commands are queued and processed reliably in the expected order.	Pass	BG/ 07-24-26	
P2-VV-016	Queues	CAN bus Queue visible in authorized user GUI	Functional test / observation	Authorized user GUI displays the CAN bus queue.	1) Log in as authorized user. 2) Navigate to CAN bus queue view. 3) Generate/observe CAN messages. 4) Confirm queue display updates.	Pass when authorized users can view current CAN bus queue information and unauthorized users cannot.	Pass	BG/ 07-24-26	